

Geometry
Unit 8
Lesson 3 Practice

Name _____
Date _____
Period _____

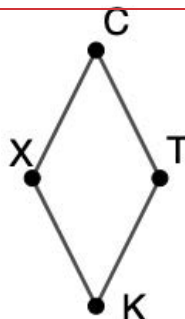
Rhombus $CTKX$ is shown where $CT = 15x - 3$ and $TK = 7x + 1$.

Commented [DB1]: <https://www.geogebra.org/calculator/zvymcdxz>

1. Select all true properties of $CTKX$.
 - ☐ Opposite sides are parallel.
 - ☐ Opposite sides are congruent.
 - ☐ Consecutive angles are congruent.
 - ☐ All angles are right angles.
 - ☐ Diagonals are perpendicular.
 - ☐ All sides are congruent.

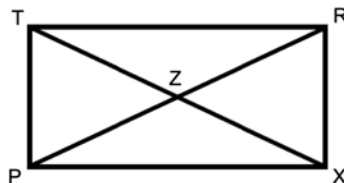
2. What is the value of x ?

3. What is the length of $\overline{CT}$?



Rectangle $TRXP$ is shown with diagonals $\overline{TX}$ and $\overline{RP}$ which intersect at point Z .

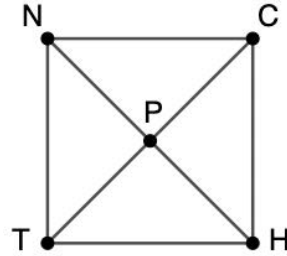
4. Write two congruent statements using the diagonals of $TRXP$.



5. $TZ = (5y - 8)$, and $ZX = (3y + 16)$. What is the value of y ?

6. What is the length of $\overline{RP}$?

Quadrilateral $NCHT$ is shown with diagonals $\overline{NH}$ and $\overline{CT}$ intersecting at point P .



7. What value of x makes $NCHT$ a square, if $m\angle TNC = 90^\circ$, $m\angle NCH = 90^\circ$, $CH = 21$, and $HT = (4x - 7)$?
8. Given $NCHT$ is a square with $PC = (5y - 2)$ and $PT = (3y + 4.74)$. What is the length of $\overline{NH}$, ?
9. What are some similarities between $CTKX$ and $NCHT$?
10. What are some differences between $TRXP$ and $NCHT$?